DLA-MatchNet for Few-Shot Remote Sensing Image Scene Classification

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***Abstract*— Few-shot scene classification aims to recognize unseen scene concepts from few labeled samples. However, most existing works are generally inclined to learn metalearners or transfer knowledge while ignoring the importance to learn discriminative representations and a proper metric for remote sensing images. To address these challenges, in this article, we propose an end-to-end network for boosting a few-shot remote sensing image scene classification, called discriminative learning of adaptive match network (DLA-MatchNet). Specifically, we first adopt the attention technique to delve into the interchannel and interspatial relationships to automatically discover discriminative regions. Then, the channel attention and spatial attention modules can be incorporated with the feature network by using different feature fusion schemes, achieving "discriminative learning." Afterward, considering the issues of the large intraclass variances and interclass similarity of remote sensing images, instead of simply computing the distances between the support samples and query samples, we concatenate the support and query discriminative features in depth and utilize a matcher to "adaptively" select the semantically relevant sample pairs to assign similarity scores. Our method leverages an episode-based strategy to train the model. Once trained, our model can predict the category of query image without further fine-tuning. Experimental results on three public remote sensing image data sets demonstrate the effectiveness of our model in the few-shot scene classification task.**

***Index Terms*— Few-shot learning, remote sensing image scene classification.**

I. Introduction

SCENE classification, as an important means of remote sensing image understanding, is to classify each remote sensing image into different scene categories according to its contents [1]–[7]. It plays an important role in various applications, including traffic control, disaster monitoring [8], [9], and land use and land cover (LULC) determination [10]–[12].

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Deep learning has achieved encouraging performance for the task of remote sensing image scene classification [13]–[17].

However, these excellent models depend heavily on a mass of labeled training samples. Once the available labeled data are insufficient, deep learning models have a risk of overfitting, resulting in performance degradation. However, in practical applications, collecting a substantial amount of annotated remote sensing images is quite labor-intensive and time-consuming. Furthermore, a trained deep learning model severely lacks the ability to recognize novel scene classes that were unseen in the training. Expanding the set of scene classes that deep learning models can identify, an abundant quantity of labeled training samples for the novel scene classes are collected. Meanwhile, a training procedure with high computational cost also needs to be restarted to avoid catastrophic forgetting.

Inspired by the rapid learning capability of humans, a new research direction emerged, which empowers machine vision systems to quickly learn novel classes from limited annotated data, referred to as few-shot learning [18], [19]. Some recent works utilize few-shot learning or transfer learning models to maintain the classification performance and meanwhile improve the generalization ability to novel categories with only a small number of labeled samples. Zhai *et al.* [20] proposed a metalearning-based remote sensing scene classification model, called LLSR, to recognize a novel class quickly under limited annotated examples.

The abovementioned method mainly focuses on learning metalearners or transfer knowledge to solve the few-shot remote sensing image scene classification problem while ignoring the importance to learn discriminative feature representations for remote sensing images [21], as shown in Fig. 1.

1) Irrelevant Object Interference: Due to the shooting in an overhead view, the remote sensing scene image inevitably contains the objects unrelated to the semantic category of the scene, which will cause adverse interference to the classification performance [22]. In Fig. 1(a), the scene category in the first image is the basketball court. The main target object is the basketball court area, but there are also unrelated objects, such as cars, buildings, and plants in the image. A similar situation appears in the rest of the images.

2) Large Intraclass Variances: Ground objects are complex and diverse, and in the process of shooting, they are also affected by background, illumination, scale, and other imaging conditions. Therefore, objects in scene images of the same

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Fig. 1. (a) Irrelevant object interference. (b) Large intraclass variances. (c) Large interclass similarity.

semantic category will be very diverse, and there may be a certain visual difference, which is the intraclass variance. Just like the images in the first row of Fig. 1(b), whose scene category is railway station, this scene has many different forms of expression. Similarly, the same is true of the scene images in the second row of Fig. 1(b).

3) Large Interclass Similarity: Scene images of different semantic categories may also have certain similarities, viz., different scene semantic categories may also contain similar ground object types with slight differences in spatial layout and density distribution. As shown in Fig. 1(c), "commercial" and "dense residential" are almost indistinguishable, which is the large interclass similarity. In view of the abovementioned factors, extracting the discriminative features of remote sensing images is significantly important to the success of few-shot remote sensing image scene classification.

In addition, most few-shot learning methods directly calculate the distances between the support samples and query samples and learn a classifier according to the distances [23], [24]. Due to the issues of the large intraclass variances and interclass similarity of remote sensing images, this may cause classification confusion, resulting in scene images that are not of the same class being divided into one category and vice versa. Therefore, learning a classifier that can adapt to remote sensing images plays an important role in improving the performance of scene classification.

To this end, in this article, we propose a novel scene classification model of remote sensing images that not only provides few-shot learning ability but also remedies the aforementioned disadvantages, termed discriminative learning of adaptive match network (DLA-MatchNet). First, we introduce the attention mechanism into the feature learning process to guide our model to focus on discriminative regions. Specifically, we design a novel attention learning module that contains two different branches: one is the spatial attention branch, and the other is the channel attention branch. The two branches learn the attention features in a parallel manner. For the channel attention branch, we perform sequentially squeeze and aggregation operations on the input feature and generate a channel attention map. This channel attention map can be viewed as a weight distribution map of the corresponding feature maps, which helps the classification model to emphasize valuable feature information. For the spatial attention branch, we aim at learning a spatial attention map that refers to the spatial position of the feature maps that should be concerned. Channel attention focuses on "what" is the scene semantic associated with given an input image, whereas spatial attention focuses on "where" is the discriminative information, both supplement each other. After that, the channel attention and spatial attention branches can be incorporated with the feature network by using different feature fusion schemes, achieving "discriminative learning." Second, we further utilize an adaptive matcher that is a multilayer perceptron (MLP) to measure the similarity scores between the images. Thereinto, "adaptive" matcher means that we no longer directly calculate the distance between samples but concatenate the support and query discriminative features in depth and then feed the combined feature maps into matcher, making this MLP to "adaptively" choose the semantically relevant sample pairs to assign similarity scores. The proposed method adopts an episode-based strategy to train an end-to-end classification model. After trained, our model can predict the category of query image without further fine-tuning. Experimental results on three public remote sensing image data sets demonstrate the effectiveness of our model in the few-shot scene classification task.

In summary, the main contributions of our proposed model can be summarized as follows.

1) By studying the interchannel and the interspatial relation, a novel feature learning module with the attention mechanism is proposed to learn robust and discriminative feature representations for few-shot remote sensing scene images.
2) We further utilize an adaptive matcher to measure similarity scores among samples instead of the handcrafted-fixed metrics (e.g., Euclidean distance). Through the learnable process of the matcher, it can adaptively deal with the problems of intraclass difference and interclass similarity.
3) Experimental results on three state-of-the-art remote sensing image data sets show the compelling performance of our model in the scene classification task.

The remaining of this article is organized as follows. Some related works are discussed in Section II. In Section III, we describe the proposed method in detail. Section IV shows the experimental results on several public remote sensing data sets. Finally, Section V presents the conclusion we draw.

II. Related Works

In this section, we will review the related works about remote sensing image scene classification and few-shot learning.

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A. Remote Sensing Image Scene Classification

In recent years, deep learning has been successfully applied in various fields due to its strong learning ability [25]–[29], hence attracting tremendous studies to address the problem of remote sensing image scene classification [30], [31]. For example, Lu *et al.* [32] proposed a multisource compensation network to solve the issues of not capturing the distribution for each source domain, thus achieving cross-domain scene classification. Lu *et al.* [33] proposed a feature aggregation CNN to learn the semantic label information for improving scene classification. Likewise, Nogueira *et al.* [34] treated CNN as a feature extractor and linear SVM as a classifier to perform remote sensing image scene classification. Cheng *et al.* [35] proposed discriminative CNNs (D-CNNs) to solve the issues of intraclass diversity and interclass similarity. Despite effective, most of these models only consider feature maps from the holistic perspective while neglecting to mine the discriminant information from each dimension of the feature maps. Different from them, our work tries to enhance the feature representation ability of our few-shot model and extract more critical feature information by utilizing comprehensively channelwise and spatialwise feature maps.

B. Few-Shot Learning

Most existing excellent algorithms adopt metalearning to solve the few-shot learning problem [36]–[40]. Recently, few-shot learning has made outstanding progress in the field of remote sensing images. For instance, Gao *et al.* [41] and Liu *et al.* [42] proposed some effective few-shot learning methods for hyperspectral image (HSI) classification. Rostami *et al.* [43] presented a few-shot synthetic aperture radar (SAR) image classification model. Among these methods, some adopt transfer learning and some adopt metalearning. The goal of transfer learning is to extract some knowledge from source tasks and apply it to a similar target task. Metalearning aims to enable machines to learn certain knowledge, patterns, or properties of algorithms from past tasks, and to flexibly and quickly guide the learning of new tasks. Although the experimental results of the above works have proved that exploiting transfer learning can improve the classification accuracy from limited labeled samples, some literature works [44] have been shown that transfer learning does not work well when the target task is not so similar to the source tasks, whereas metalearning, such as human beings, can "draw inferences about other cases from one instance," so it is more flexible than transfer learning. Therefore, we adopt the metalearning-based method to address the few-shot problem. These methods mentioned earlier lack the ability to learn more discriminant features in the face of complex remote sensing images. They also fail to consider learning a reliable and robust metric specifically applicable for remote sensing images.

III. Proposed Method

A. Overview of the Proposed Method

We use the popular few-shot learning for remote sensing image scene classification, referred to as DLA-MatchNet, where a feature learning module acquires the remote sensing images features, and then, a matcher calculates the similarity scores between sample features. As shown in Fig. 2, on the one hand, DLA-MatchNet applies the attention mechanism to the feature learning module to make the model adaptively learn the feature information related to category semantics by discriminative learning. On the other hand, instead of using the predefined linear metrics, our model adopts an adaptive flexible metric learning scheme to perform classification.

In our work, there are three data sets: training set D_{train}, validation set D_{val}, and test set D_{test}, and they have their own label space, disjoint with each other, i.e., $L_{\text{train}} \cap L_{\text{val}} = \emptyset$, $L_{\text{train}} \cap L_{\text{test}} = \emptyset$ and $L_{\text{val}} \cap L_{\text{test}} = \emptyset$. The role of the validation set D_{val} is to select the parameters corresponding to the model with the best performance. Due to the few-shot learning setting, we have to train models with only few training examples. However, using such few samples to train the model inevitably leads to overfitting, and thus, we perform metalearning on the training set and extract metaknowledge to help the model better perform few-shot learning on the test set.

We adopt an episode-based training manner on the training set to imitate the few-shot learning setting. In the same way, for the testing phase, we also need to split the test set into many episodes. Formally, in the single training phase of our method (DLA-MatchNet), an episode contains a support set D_{spt} and a query set D_{qry}. Concretely, C classes are randomly chosen from the training set, and K labeled samples of each class are selected as the support set $D_{\text{spt}} = (x_i, y_i | i = 1, \ldots, m)$ $(m = C \times K)$. Then, a portion of the remaining samples from these categories are used to form the query set $D_{\text{qry}} = \left(x_j, y_j | j = 1, \ldots, n\right)$. The model learns how to distinguish between these C classes from $C \times K$ samples, which is called C-way K-shot.

B. Feature Learning

We introduce the attention mechanism into the feature learning module to study the interchannel and the interspatial relation to learn more discriminative feature representations.

Suppose that we want to obtain the attention-aware feature of the lth layer, where this attention-aware feature is synthesized by combining the original feature with spatial and channel attention maps. Formally, the input feature of the lth layer is $\mathbf{X}^{l-1} \in \mathbb{R}^{c \times h \times w}$, and the feature learning module processes $\mathbf{X}^{l-1}$ by using spatial and channel attention maps together as

$$\begin{aligned} \mathbf{Z}^l &= \text{CNN}_{\text{layer_}l}\left(\mathbf{X}^{l-1}\right) \\ \gamma^l &= \mathbf{\Phi}\left(\mathbf{Z}^l\right) \\ \mathbf{X}^l &= f\left(\mathbf{Z}^l, \gamma^l\right) \end{aligned} \tag{1}$$

where the feature map $\mathbf{Z}^l$ is the output of the lth convolution layer and f is the feature fusion function that integrates the original feature $\mathbf{Z}^l$ and the attention maps effectively. In DLA-MatchNet, there are two feature fusion schemes, which will be described in detail later. $\mathbf{\Phi}$ is an attention function, which is used to calculate the spatial attention and channel attention maps. Next, we will describe the specific process of generating

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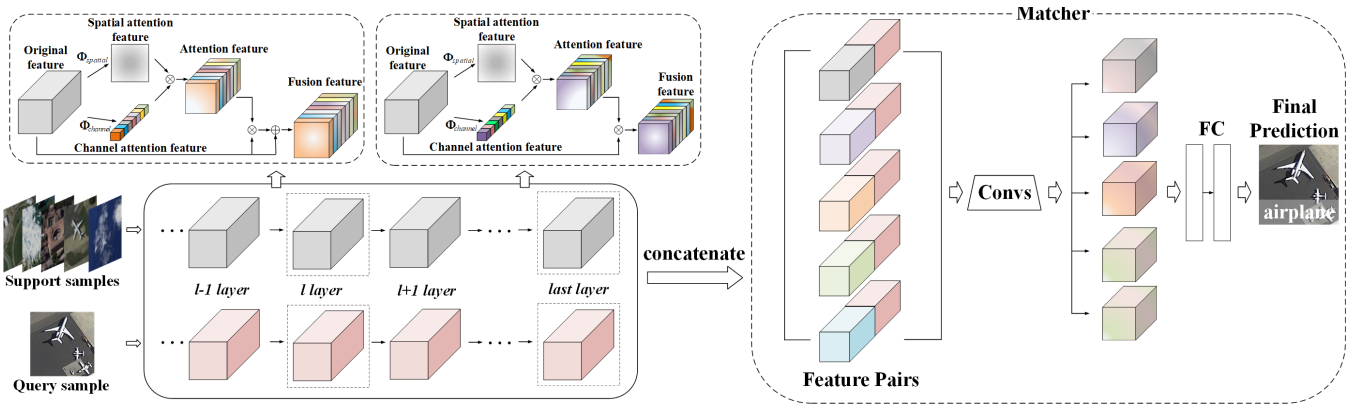


Fig. 2. Overall architecture of our proposed model for a five-way one-shot setting with one query image.

the spatial attention and channel attention maps in detail in the following.

First, let $\mathcal{S}_{\text{sp}}$ represent a spatial attention map, which refers to the spatial position of the feature maps that should be concerned. Spatial attention takes each pixel of the feature map as a unit and assigns a weight to each pixel of the feature map. To acquire the spatial attention map $\mathcal{S}_{\text{sp}}$, we conduct an average-pooling operation along the channel axis to generate a spatial descriptor. On the spatial descriptor, we use a 3×3 convolution layer of stride 2 and an upsampling layer in succession. Then, through a convolution layer of 1×1 [45], we get the spatial attention map $\mathcal{S}_{\text{sp}}$. The function of this 1×1 convolution layer is to further learn the appropriate spatial attention map so that it can be adaptively combined with the channel attention map. Formally, let $\mathbf{\Phi}_{\text{sp}}$ denote the spatial attention function and $\mathbf{Z} \in \mathbb{R}^{c \times h \times w}$ stand for the original feature that is not processed by the attention mechanism. The spatial descriptor $\mathbf{F}_s^{\text{avg}} \in \mathbb{R}^{1 \times h \times w} = \text{AvgPool}_{\text{sp}}(\mathbf{Z})$ of $\mathbf{Z}$ is obtained by utilizing the average-pooling operation. Next, $\mathbf{F}_s^{\text{avg}}$ is convolved via a convolution layer followed by an upsampling operation and a convolution operation. The details are as follows:

$$\mathcal{S}_{\text{sp}} = \text{Conv}^{1\times1}\big(\text{Upsampling}\big(\text{Conv}^{3\times3}\big(\text{AvgPool}_{\text{sp}}(\mathbf{Z})\big)\big)\big) \quad (2)$$

where $\text{Conv}^{1\times1}$ (or $\text{Conv}^{3\times3}$) is a 1×1 (or 3×3) convolution operation with a ReLU function. In addition, (2) can also be written as

$$\mathcal{S}_{\text{sp}} = \mathbf{\Phi}_{\text{sp}}(\mathbf{Z}). \quad (3)$$

Second, let $\mathcal{S}_{\text{ch}}$ be a channel attention map, which takes each feature map as a unit and assigns a weight to each channel, and the weights should be a vector. Each channel of feature maps represents a semantic attribute, and channelwise attention map $\mathcal{S}_{\text{ch}}$ can be regarded as the weights assigned to the semantic attributes. We perform an average-pooling operation to aggregate the spatial information of the original feature $\mathbf{Z}$, thus producing a channel descriptor that can be viewed as each channel response to the entire original feature $\mathbf{Z}$. This channel descriptor is insufficient to reflect the relation between feature channels. To provide a completer and more accurate interchannel relation, we further perform two convolution operations on this channel descriptor. Formally, let $\mathbf{\Phi}_{\text{ch}}$ denote the channel attention function. The channel descriptor $\mathbf{F}_c^{\text{avg}} \in \mathbb{R}^{c\times1\times1} = \text{AvgPool}_{\text{ch}}(\mathbf{Z})$ of $\mathbf{Z}$ is obtained by utilizing the average-pooling operation. Then, $\mathbf{F}_c^{\text{avg}}$ is successively fed into two 1×1 convolution layer with ReLU function. The specific operation is as follows:

$$\mathcal{S}_{\text{ch}} = \text{Conv}_2^{1\times1}\big(\text{Conv}_1^{1\times1}\big(\text{AvgPool}_{\text{ch}}(\mathbf{Z})\big)\big) \quad (4)$$

where the parameter matrix of $\text{Conv}_1^{1\times1}$ is different from that of $\text{Conv}_2^{1\times1}$. To reduce the number of the model parameters, we set the parameter matrices of $\text{Conv}_1^{1\times1}$ and $\text{Conv}_2^{1\times1}$ as $W_1 \in \mathbf{R}^{(c/r)\times c}$ and $W_2 \in \mathbf{R}^{c\times(c/r)}$, respectively. We also set r as 16 in our experiments. Likewise, (4) can also be written as

$$\mathcal{S}_{\text{ch}} = \mathbf{\Phi}_{\text{ch}}(\mathbf{Z}). \quad (5)$$

The spatial attention map $\mathcal{S}_{\text{sp}}$ and channel attention map $\mathcal{S}_{\text{ch}}$ are combined to enhance the discriminative feature representations of our model. Thus, we generate the attention map $\mathcal{S}_{\text{att}} \in \mathbf{R}^{c\times h\times w}$ by the following equation:

$$\mathcal{S}_{\text{att}} = \text{Sigmoid}\big(\text{Conv}\big(\mathcal{S}_{\text{sp}} \times \mathcal{S}_{\text{ch}}\big)\big) \quad (6)$$

where Conv stands for a convolutional block, which includes a 1×1 convolution layer, a batch normalization, and a ReLU nonlinearity layer.

In CNN, the low-level network can extract more low-level feature details, whereas the high-level network can extract global semantic information. Only when the low-level network extracts more feature details, the high-level features can better abstract the scene object that should be paid attention to. Therefore, we not only add the attention mechanism into the last layer of the feature learning module but also apply the attention mechanism into the earlier layers of our feature learning module. In addition, the feature fusion scheme of the earlier layers is different from that of the last layer.

For the first feature fusion scheme, suppose that the output original feature of the earlier layers is $\mathbf{Z}_{\text{ear}} \in \mathbf{R}^{c\times h\times w}$. After processing by the attention mechanism, we can obtain

$$\mathcal{S}_{\text{att}}^{\text{ear}} = \text{Sigmoid}\big(\text{Conv}\big(\mathbf{\Phi}_{\text{sp}}(\mathbf{Z}_{\text{ear}}) \times \mathbf{\Phi}_{\text{ch}}(\mathbf{Z}_{\text{ear}})\big)\big). \quad (7)$$

Then, integrating the original feature $\mathbf{Z}_{\text{ear}}$ and the attention map $\mathcal{S}_{\text{att}}^{\text{ear}}$ through the following equation:

$$\mathbf{X}_{\text{ear}} = \big(1 + \mathcal{S}_{\text{att}}^{\text{ear}}\big) \times \mathbf{Z}_{\text{ear}} \quad (8)$$

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where the elements of $\mathcal{S}_{\text{att}}^{\text{ear}}$ range from [0, 1], so $\mathbf{X}_{\text{ear}}$ just enhances some of the key information but is still approximate to $\mathbf{Z}_{\text{ear}}$. This feature fusion scheme is applicable to the earlier layers of the model, which can avoid the influence of the feature value degradation brought by the general dot product operation and facilitate the feature operation of the subsequent layer.

For the second feature fusion scheme, suppose that the output original feature of the last layers is $\mathbf{Z}_{\text{last}} \in \mathbf{R}^{c\times h\times w}$. Before performing the attention mechanism operation, we input $\mathbf{Z}_{\text{last}}$ into the spatial transformer network (STN) [46] module to enhance the spatially invariant ability of our model and further improve the classification performance, denoted as $\mathbf{Z}_{\text{last}}^{\text{STN}} \in \mathbf{R}^{c\times h\times w}$. Then, we can obtain the attention map $\mathcal{S}_{\text{att}}^{\text{last}} \in \mathbf{R}^{c\times h\times w}$ by using the attention mechanism

$$\mathcal{S}_{\text{att}}^{\text{last}} = \text{Sigmoid}\big(\text{Conv}\big(\mathbf{\Phi}_{\text{sp}}\big(\mathbf{Z}_{\text{last}}^{\text{STN}}\big) \times \mathbf{\Phi}_{\text{ch}}\big(\mathbf{Z}_{\text{last}}^{\text{STN}}\big)\big)\big). \tag{9}$$

Then, integrating the feature $\mathbf{Z}_{\text{last}}^{\text{STN}}$ and the attention map $\mathcal{S}_{\text{att}}^{\text{last}}$ through the following equation:

$$\mathbf{X}_{\text{last}} = \mathcal{S}_{\text{att}}^{\text{last}} \times \mathbf{Z}_{\text{last}}^{\text{STN}}. \tag{10}$$

Because this second scheme performs feature fusion at the last layer of the feature learning module, the dot product operation will not affect the feature learning performance.

C. Matcher Learning

We utilize a learnable and adaptive matcher, which is an MLP, for remote sensing images [47]. In each episode, we input the support samples and the query samples into the feature learning module and get the attentionaware features of the support samples and the query samples. Instead of directly computing the distances between the support features and query features, we concatenate the support and query discriminative features in depth and then feed the combined feature maps into matcher, making this MLP to "adaptively" choose the semantically relevant sample pairs to assign more similarity scores. In a nutshell, we adopt a nonlinear network structure (i.e., MLP) to measure the similarity of remote sensing image samples adaptively and abandon the handcrafted and fixed metrics (e.g., Euclidean distance), which is not suitable for complex remote sensing image classification.

Formally, in the C-way K-shot setting, support samples $x_i \in D_{\text{spt}}$ ($i = 1, 2, \ldots, m$ and $m = C \times K$) and one query sample $x_j \in D_{\text{qry}}$ are fed into the feature learning module to obtain the attention-aware features $\hat{\mathbf{X}}_i$ ($i = 1, 2, \ldots, m$) and $\hat{\mathbf{X}}_j$, respectively.

For $K = 1$, inputting the attention-aware features $\hat{\mathbf{X}}_i$ and $\hat{\mathbf{X}}_j$ to our matcher g_ϕ in the form of concatenation, let $\mathcal{C}$ be a concatenation operation. Then, we can obtain classification scores s_{ij}

$$s_{ij} = g_\phi\Big(\mathcal{C}\Big(\hat{\mathbf{X}}_i, \hat{\mathbf{X}}_j\Big)\Big) \quad i = 1, 2, \ldots, C \tag{11}$$

where ϕ is the parameter of the matcher g_ϕ.

For $K > 1$, we need to sum all the attention-aware features of each class in the support set D_{spt} and obtain the class-level attention-aware feature $\sum_{k=1}^{K} \hat{\mathbf{X}}_c^k = \hat{\mathbf{X}}_c$ ($c = 1, 2, \ldots, C$). In the same way, concatenating the query attention-aware feature $\hat{\mathbf{X}}_j$ and the class-level attention-aware feature $\hat{\mathbf{X}}_c$ and inputting it into g_ϕ, hence, the classification scores are

$$s_{cj} = g_\phi\Big(\mathcal{C}\Big(\hat{\mathbf{X}}_c, \hat{\mathbf{X}}_j\Big)\Big) \quad c = 1, 2, \ldots, C. \tag{12}$$

Then, we utilize a mean square error (MSE) loss to train our model, whose specific function form can be written as

$$L\big(s_{ij}\big) = \sum_{i=1}^{C}\sum_{j=1}^{n}\big(s_{ij} - \theta\big)^2\theta = \begin{cases} 1, & \text{if } y_i = y_j \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

D. Network Architecture

Our few-shot remote sensing image scene classification model consists of two modules, namely, the feature learning module and matcher. In our method, we do not use any pretrained convolutional network model.

In the feature learning module, it consists of five convolutional blocks, and each convolutional block has a 3×3 convolution layer, a batch normalization, and a ReLU nonlinearity layer. Among them, the first four convolutional blocks contain a 2×2 max-pooling layer, whereas the last convolutional block does not. We apply the attention mechanism between the third convolutional block and the fourth convolutional block and, meanwhile, adopt the first feature fusion scheme to integrate the original feature with the spatial and channel attention feature maps. In the last layer of our feature learning module, we use STN and the attention mechanism in sequence to further enhance the feature learning ability of the model and make use of the second feature fusion scheme to produce attention-aware features.

Compared with the feature learning module, the structure of matcher is relatively simple, mainly including two convolution blocks and two full connection layers. Thereinto, the two convolution blocks both contain a 3×3 convolution layer, a batch normalization, a ReLU nonlinearity layer, and a 2×2 max-pooling layer. The output of the first fully connected (FC) layer is calculated by ReLU, whereas the output of the last FC layer is calculated by using Sigmoid.

IV. Experiments

In this section, we perform experiments to evaluate the effectiveness of our proposed method.

A. Data Set Description

We verify the performance of our model on three publicly available data sets, including the NWPU-RESISC45 data set [22], WHU-RS19 data set [48], and UC Merced data set [49]. We follow the standard splits and experimental rules of the commonly used few-shot learning data sets (such as miniImageNet [24]), as shown in Table I.

The NWPU-RESISC45 data set consists of 31 500 images in a total of 45 different scene classes, each of which provides 700 images. The size of each image is 256×256 pixels in the RGB color space. In our experiments, we split the data set into 25, 10, and 10 classes for training, validation, and testing, respectively, and the exact splits are shown in Table I.

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TABLE I
EXACT SPLITS ON THREE DATA SETS

Datasets	Training	Validation	Testing
NWPU-RESISC45	Airplane; Baseball diamond; Beach; Bridge; Chaparral; Church; Cloud; Desert; Freeway; Golf course; Harbor; Island; Lake; Meadow; Mobile home park; Mountain; Palace; Railway; Rectangular farmland; Roundabout; Sea ice; Ship; Sparse residential; Stadium; Wetland;	Commercial area; Industrial area; Overpass; Railway station; Runway; Snowberg; Storage tank; Tennis court; Terrace; Thermal power station;	Airport; Basketball court; Circular farmland; Dense residential; Forest; Ground track field; Intersection; Medium residential; Parking lot; River;
WHU-RS19	Airport; Bridge; Desert; Football Field; Industrial; Mountain; Parking; Port; Residential;	Beach; Farmland; Forest; Park; Railway Station;	Commercial; Meadow; Pond; River; Viaduct;
UC Merced	Agricultural; Baseball diamond; Buildings; Chaparral; Dense residential; Freeway; Harbor; Medium residential; Overpass; Parking lot;	Airplane; Forest; Intersection; Runway; Storage tanks;	Beach; Golf course; Mobile home park; River; Sparse residential; Tennis court;

The WHU-RS19 data set is released by Wuhan University, which is also a remote sensing image data set. It contains 19 different scene classes, each with a sample size greater than or equal to 50 images. The total number of samples in this data set is 1005, and the image pixel size is 600×600. For the WHU-RS19 data set, we divide nine classes as a training set, five classes as a validation set, and five classes as a test set (see Table I for details).

The UC Merced data set is a 21-level remote sensing data set of land use images for research purposes. This data set consists of 21 scene classes, and each class has 100 images in total. The pixel size of the image is 256×256. The data set was released in 2010 by the UC Merced Computer Vision Laboratory. In this data set, ten classes are selected as a training set, six classes are regarded as a validation set, and five different classes are regarded as a test set, as shown in Table I.

B. Parameter Setting

We implement our model in the PyTorch [50] framework, and GPU is NVIDIA GeForce GTX. All remote sensing images are resized to 256×256. For the feature learning module, we set the output channels of the first three convolutional blocks as $c_1 = 32$, $c_2 = 64$, and $c_3 = 128$, respectively. The output channels of the last two convolutional blocks are the same, i.e., $c_4 = c_5 = 256$. For the matcher, we set the output channels of the first two convolutional blocks to be 256 and 256. The input dimensions of two FC layers are $256 \times 2 \times 2$ and 32, whereas the output dimensions are 32 and 1. We adopt ADAM to optimize our model at the initial learning rate 1×10^{-4}. In the localization network of STN, the bias of the last full connection layer is set as vector $[-0.75, 0, -0.5, 0, -0.75, 0.75]$.

C. Experimental Results and Comparisons

We adopt the overall accuracy, which is defined as the ratio of all correctly predicted images to the total number of images, as evaluation metrics in our experiments. In addition to LLSR, we also adopt several classical few-shot learning methods for experimental comparison, including PN [23], MatchingNet [24], RN [47], MAML [51], and Meta-SGD [52]. Note that we set the feature extraction network of RN as five convolutional blocks in our experiments. PN, MatchingNet, and RN belong to the metric learning-based approach. PN and MatchingNet adopt predefined and fixed metrics, whereas RN uses a nonlinear comparator. MAML, Meta-SGD, and LLSR need to fine-tune on the few-shot problem during the test phase. Tables II–IV show the experimental results of these methods on three data sets, respectively. The experimental results of LLSR are borrowed from their report.

TABLE II
FEW-SHOT CLASSIFICATION ACCURACIES ON THE NWPU-RESISC45 DATA SET

Method	5-way	
	1-shot	5-shot
PN [23]	$40.33 \pm 0.18\%$	$63.82 \pm 0.56\%$
MatchingNet [24]	37.61%	47.10%
MAML [51]	$48.40 \pm 0.82\%$	$62.90 \pm 0.69\%$
Mate-SGD [52]	$60.63 \pm 0.90\%$	$75.75 \pm 0.65\%$
LLSR [20]	51.43%	72.90%
RN [47]	$66.43 \pm 0.73\%$	$78.35 \pm 0.51\%$
DLA-MatchNet	$68.80 \pm 0.70\%$	$81.63 \pm 0.46\%$

The experimental results shown in Table II are calculated by averaging on 1000 randomly generated episodes from the test data set. From Table II, in the case of five-way one-shot, DLA-MatchNet achieves the highest accuracy of 68.80%, surpassing the second best RN (without attention) by 2.37% in the overall accuracy. This suggests the importance of discriminative learning in remote sensing image scene classification and the effectiveness of our attention learning approach in a more challenging few-shot scenario. The experimental results under the case of five-way five-shot show that DLA-MatchNet again outperforms all compared methods with obvious accuracy advantages. Importantly, DLA-MatchNet (without fine-tuning) is superior to Mate-SGD (needs fine-tuning) by a large margin.

In addition, we also conduct comparative experiments on the WHU-RS19 data set, and the experimental results are computed averagely on 100 randomly generated episodes from

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TABLE III
FEW-SHOT CLASSIFICATION ACCURACIES ON THE WHU-RS19 DATA SET

Method	5-way	
	1-shot	5-shot
PN [23]	58.01±0.16%	80.70±0.11%
MatchingNet [24]	50.13%	54.10%
MAML [51]	49.13±0.65%	62.49±0.51%
Mate-SGD [52]	51.54±2.31%	61.74±2.02%
LLSR [20]	57.10%	70.65%
RN [47]	60.92±1.86%	79.75±1.19%
DLA-MatchNet	68.27±1.83%	79.89±0.33%

TABLE IV
FEW-SHOT CLASSIFICATION ACCURACIES ON THE UC MERCED DATA SET

Method	5-way	
	1-shot	5-shot
PN [23]	52.27±0.20%	69.86±0.15%
MatchingNet [24]	34.70%	52.71%
MAML [51]	48.86±0.74%	60.78±0.62%
Mate-SGD [52]	50.52±2.61%	60.82±2.00%
LLSR [20]	39.47%	57.40%
RN [47]	48.08±1.67%	61.88±0.50%
DLA-MatchNet	53.76±0.62%	63.01±0.51%

the test data set. As shown in Table III, in the case of five-way one-shot, DLA-MatchNet achieves the highest recognition accuracy of 68.27%, which is 7.35% higher than that of RN (without attention). For the five-way five-shot settings, we can see that DLA-MatchNet achieved competitive results. DLA-MatchNet achieves a small margin (+0.14%) over RN (without attention). The highest accuracy of 80.70% is achieved by PN (using Euclidean distance), which is 0.81% higher than DLA-MatchNet. However, it is worth pointing out that our algorithm is superior to PN by 10.26% in the case of five-way one-shot with fewer samples that are more challenging.

For the UC Merced data set, from the experimental results in Table IV, we can see that our method achieved state-of-the-art results on five-way one-shot and competitive performance on five-way five-shot. For example, DLA-MatchNet achieves 53.76% (one-shot) and 63.01% (five-shot), which are higher than RN (without attention). Our method (without fine-tuning) also performs better than MAML, Meta-SGD, and LLSR. DLA-MatchNet is superior to MatchingNet (using cosine distance) but lower than PN (using Euclidean distance) by 6.8% (five-shot). However, in the more rigorous one-shot case, DLA-MatchNet is better than PN.

In conclusion, first, our method achieves discriminative feature learning in the few-shot remote sensing scene classification task, whereas these comparison methods only adopt the simple feature extraction networks that have only four convolution layers to learn image features. The experimental results also strongly verified the importance of discriminative learning in remote sensing image scene classification and the effectiveness of our attention learning module in the few-shot scenario. Second, we propose a learnable and adaptive matcher, which is an MLP, to solve the issues of the large intraclass variances and interclass similarity of remote sensing images. The final experimental results also confirmed that our matcher is more suitable for the few-shot remote sensing scene classification task than the predefined and fixed metrics. In addition, our method does not need to fine-tune on few-shot problems during the test phase, which is simpler and more efficient than MAML, Meta-SGD, and LLSR.

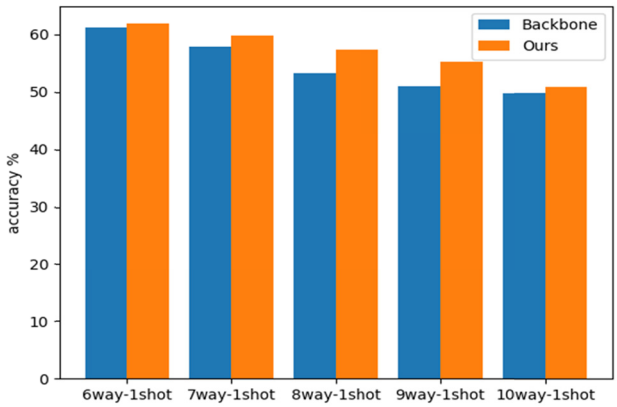


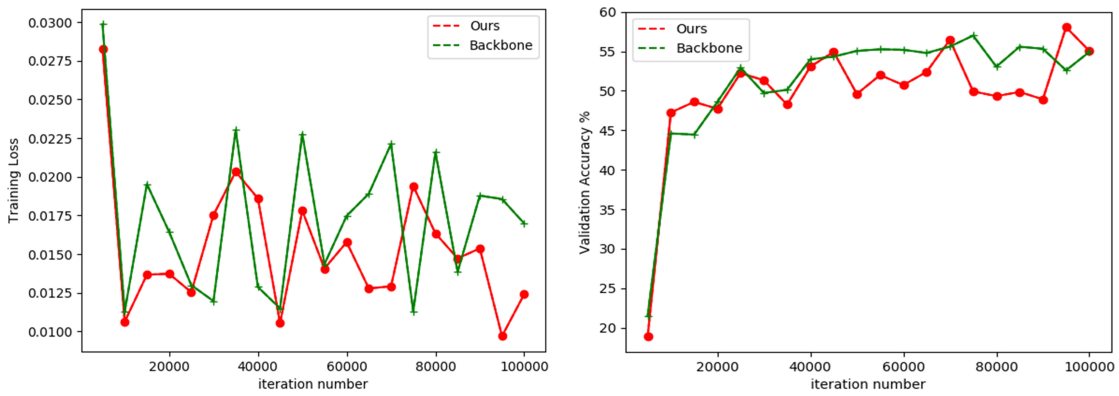
Fig. 3. Classification results of different novel classes on the NWPU-RESISC45 data set.

D. Analysis of Our Model Framework

In order to enhance the feature representation capability of the model for remote sensing images, we introduce the attention mechanism in the feature learning stage. Our attention module is conceptually similar to CBAM [53] because both are designed to learn attention features from channels and spatial dimensions. However, they differ markedly in design. First, in multiple-layer attention feature learning, CBAM only adopts a form of summation to fuse the attention maps and sample features, whereas our method uses different feature fusion schemes at different layers. Second, in the channel attention module, CBAM uses the MLP that contains two FC layers, whereas our method adopts two convolution layers with kernel 1×1 to process feature maps. In addition, the arrangement of attention modules in CBAM is a sequential channel–spatial manner, while our combination of the spatial and channel attention modules is in a parallel manner. Table V shows the experimental results on different attention arranging schemes, including CBAM. Note that in addition to the different structure of attention, other network structures remain the same and the experimental setup is consistent. Backbone does not contain any attention mechanisms and only has a feature learning module and matcher. The attention structures in the Backbone +channel+spatial and Backbone+spatial+channel methods are the same as our DLA-MatchNet, except for different arrangements. We can observe that the experimental effects of our designed attention modules are better than that of CBAM in both sequential and parallel manners.

Furthermore, in order to investigate the impact of the attention mechanisms, we added on scene classification performance and conducted ablation studies of our model framework

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(a) (b)

Fig. 4. Training loss and validation accuracy under the five-way one-shot setting on the NWPU-RESISC45 data set. (a) Training loss. (b) Validation.

TABLE V

ARRANGING MANNERS OF CHANNEL AND SPATIAL ATTENTION. THE COMPARISON EXPERIMENTS ARE PERFORMED ON THE NWPU-RESISC45 DATA SET

Method	5-way	
	1-shot	5-shot
Backbone+CBAM	67.14 ± 0.69%	79.97 ± 0.58%
Backbone+channel+spatial	67.62 ± 0.71%	81.57 ± 0.19%
Backbone+spatial+channel	67.66 ± 0.81%	80.61 ± 0.48%
DLA-MatchNet	68.80 ± 0.70%	81.63 ± 0.46%

TABLE VI

ABLATION OF OUR MODEL ON THE NWPU-RESISC45 DATA SET

Method	5-way	
	1-shot	5-shot
Backbone+A1	66.60 ± 0.69%	80.17 ± 0.48%
Backbone+A1+A2	67.16 ± 0.68%	79.58 ± 0.50%
Backbone+A1+STN	67.91 ± 0.69%	79.30 ± 0.45%
Backbone+A2	66.17 ± 0.71%	80.00 ± 0.48%
Backbone+A2+STN	67.76 ± 0.69%	80.63 ± 0.46%
Backbone+STN	67.17 ± 0.70%	80.28 ± 0.48%
Backbone	66.43 ± 0.73%	78.35 ± 0.51%
DLA-MatchNet	68.80 ± 0.70%	81.63 ± 0.46%

on the NWPU-RESISC45 data set in this section. Specifically, we consider the following scenarios:

1) *Backbone:* Regardless of the attentional mechanisms we introduced, the backbone of our model contains a feature learning module and matcher.
2) *Backbone+A1:* We only add the attention mechanism $\mathcal{A}_1$ on the basis of Backbone and adopt the first feature fusion scheme to obtain the attention-aware feature.
3) *Backbone+A1+A2:* We add the attention mechanism $\mathcal{A}_2$ on the basis of Backbone+A1 and meanwhile adopt the second feature fusion scheme to obtain the attention-aware feature.
4) *Backbone+A1+STN:* We add an STN on the basis of Backbone+A1.
5) *Backbone+A2:* We only add the attention mechanism $\mathcal{A}_2$ on the basis of Backbone and use the second feature fusion scheme to obtain the attention-aware feature.
6) *Backbone+A2+STN:* We add an STN on the basis of Backbone+A2.
7) *Backbone+STN:* We only insert an STN after the fifth convolutional block. The detailed experimental results are shown in Table VI.

From the experimental results, we can see that only by combining the three modules, namely, attention mechanism $\mathcal{A}_1$, STN, and attention mechanism $\mathcal{A}_2$, can we achieve a satisfactory classification effect. Besides, in the case of five-way one-shot, the combination of any attention selection network and STN achieves a relatively high classification accuracy. The classification accuracy of Backbone+A1+STN and Backbone+A2+STN are 67.91% and 67.76%, respectively, higher than Backbone+A1+A2, Backbone+A1, Backbone+A2, and Backbone+STN, while in the case of five-way five-shot, Backbone+A2+STN achieves 80.63% classification accuracy, which is 0.35% higher than Backbone+STN. The results of other ablation experiments are all lower than Backbone+A2+STN.

We learn a nonlinear adaptive metric (i.e., our matcher) to replace the fixed-linear metric. In Table II, whether in five-way one-shot or five-way five-shot, the experimental results strongly prove the effectiveness of the matcher on a large-scale data set. In Tables III and IV, our method achieves competitive performance under the five-way five-shot setting, except that PN has a higher performance than our algorithm. However, in the case of five-way one-shot, our algorithm is superior to PN and other algorithms. Moreover, five-way one-shot with fewer samples is more challenging than five-way five-shot, which also proves that the matcher is effective in the stricter few-shot (i.e., one-shot) remote sensing scene classification. It is worth mentioning again that under the NWPU-RESISC45 data set with various categories and more complicated-varied samples, our method is far superior to PN in both five-way one-shot and five-way five-shot cases.

To verify the classification performance of our method on different numbers of new categories, we perform several one-shot experiments on the NWPU-RESISC45 data set. Fig. 3 shows the experimental results. We can see that the classification performance of our method is always better than that

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of Backbone. Fig. 4 shows the training loss and verification accuracy of our method and Backbone. The iteration number changes from 5000 to 100 000 with a stride of 5000. As shown in Fig. 4, during the training phase, the training loss of our method is lower than that of Backbone, and the highest verification accuracy of our method is higher than that of Backbone.

V. Conclusion

This article proposes a novel scene classification model of remote sensing image that provides few-shot learning ability, called DLA-MatchNet. Our method mainly includes a feature learning module and matcher. First, we introduce the attention mechanism into the feature learning module to boost the feature representation power of DLA-MatchNet, making our model to learn more discriminative feature representations, achieving "discriminative learning." Second, we adopt an adaptive matcher to more robustly measure the similarity scores between remote sensing images. We conduct comparative experiments on several publicly remote sensing image data sets, and the experimental results show the excellent performance of DLA-MatchNet in scene classification tasks. Finally, as mentioned in [54], it is an urgent need for fusing various remote sensing data sources to avoid the waste of remote sensing data. Therefore, in our future work, we will focus on how to reasonably embed unannotated samples from the different remote sensing data sources into the few-shot learning framework to further improve the model performance.

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